

Last time: Proved $A = \{a^n b^n \mid n \geq 0\}$ is nonregular.

Today: Generalizing this approach.

How did we do it last time?

1. Assumed you were given a DFA with k states for A .
2. Picked a long string in A and a portion to target or pigeonhole principle.

Ex: $k = 7$.

$a^k b^k$. Portion to target? $y = a^k$, (Let $z = b^k$)

3. Pigeonhole principle: $\hat{\delta}(s, a^i) = \hat{\delta}(s, a^j)$ for some $i < j \in \{0, \dots, k\}$.

Exists a loop v such that $y = uvw$.

4. Generate an accepted string not in A .

Which string? uv^2wz

Today: Replicate this on a couple examples, state as a general theorem.

An Example

Let $A = \{a^\ell b^m c^n \mid m \leq \ell, n\}$

Warmup: Which strings are in A ?

- abc
- $aabbbcc$
- $aaaaccbb$
- $aaaaaaabbbbbbbcccccc$

Replicating the approach:

1. Assume you were given a DFA with k states which accepts all strings in A .
2. Pick a long string in A , and a portion to target.

Ex: $k = 7$.

$a^k b^k c^k$. Portion to target: $y = b^k$. (Let $x = a^k$, $z = c^k$)

3. Pigeonhole principle: The DFA will loop while reading y .

$$\hat{\delta}(s, xb^i) = \hat{\delta}(s, xb^j) = q, i < j \in \{0, \dots, k\}$$

Let $b^i = u$, $b^{j-i} = v$, $b^{k-j} = w$. $y = uvw$.

Any number of copies of v can be “pumped in.”

4. All of these will be accepted. Which of the following are in A ?

- $xuwz$
- $xuvwz$
- xuv^2wz
- xuv^3wz

Conclusion: If M accepts all strings in A , it also accepts other stuff.

The Pumping Lemma

Idea: State the approach as generally as possible.

Let A be a language. Assume you can do the following:

1. Let k be given. (Number of states in DFA)
2. Pick $xyz \in A$ with $|y| \geq k$. (y is targeted for pigeonhole)
3. Let $uvw = y$ be given, with $|v| \geq 1$ (v is the loop)
4. Pick $i \geq 0$ such that $xuv^i wz \notin A$.

Then A is not regular.

Interpretation: Games with a demon.

Winning strategy $\Rightarrow A$ not regular.

Another Example

$$A = \{a^{2^n} \mid n \geq 0\}$$

List the first few smallest elements of A .

Running example: $k = 4$, a^8

Game with the demon:

1. Let k be given. (Number of states in DFA)
2. Pick $a^{2^{k+1}}$. Pigeonhole window: $y =$ first k as .

What are x and z ?

3. Let $uvw = y$ be given, with $|v| \geq 1$. (We're in the same state before and after v .)

Let $v = a^j$, for $j \leq k$.

4. Question: How long is xuv^2wz ?

$$2^{k+1} + j \leq 2^{k+1} + k < 2^{k+2}.$$

Hence, A is not regular.